

Topological Simplification of Signals for Inference and Approximate Reconstruction

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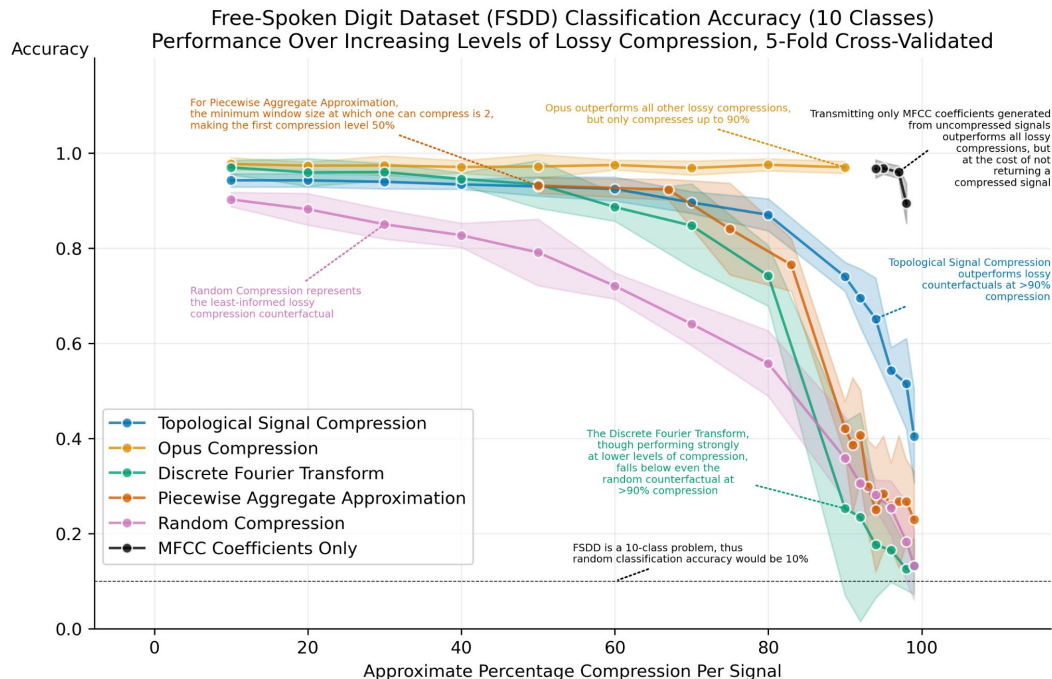
March 9th, 2023





Takeaways

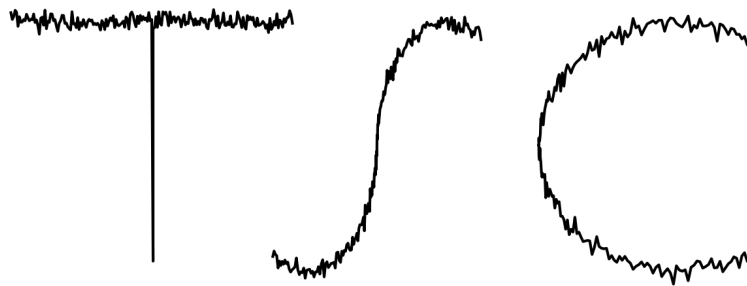
- Topology-inspired lossy compression algorithm: Topological Signal Compression (TSC).
- Generalizable to any 1d signals / sequences of floating point values.
- Interpretable.
- Robust to variable bandwidth constraints.



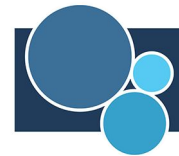


Code

- `topological-signal-compression`
[Python package](#)
- [Documentation](#)
 - [FSDD Example](#)
 - [Music \(Chopin\) Example](#)

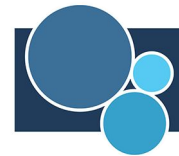


Topological Signal Compression



Most Promising TSC Use Case

- Internet of Things (IoT) with multimodal sensing capabilities.
 - Generalizability helpful.
- Avoiding black box concerns with highly-interpretable compression.
- Small bandwidth constraint requiring high levels of compression.
 - GDA's use case: 300 byte message limit with Iridium.
- Variable bandwidth constraint.
 - Relative prioritization of different modalities over time, either via automation or user requests.
 - Needs to be stable to downstream ML.
- Preserving exact values of critical points.
 - Desire to send data points rather than summary statistics or featurizations.



Overview

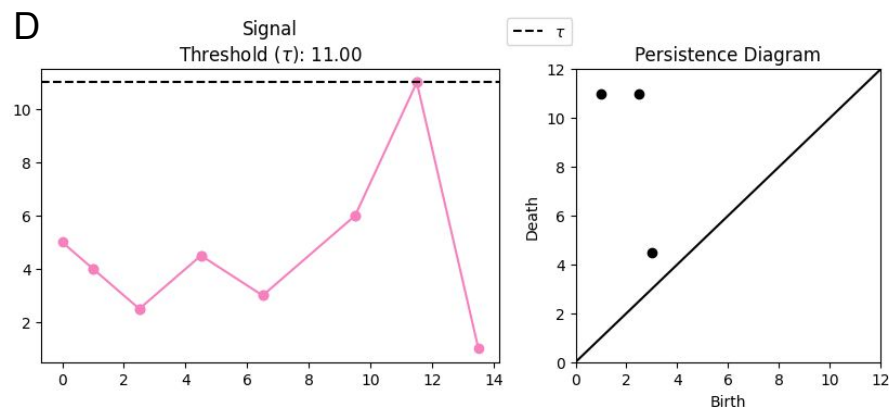
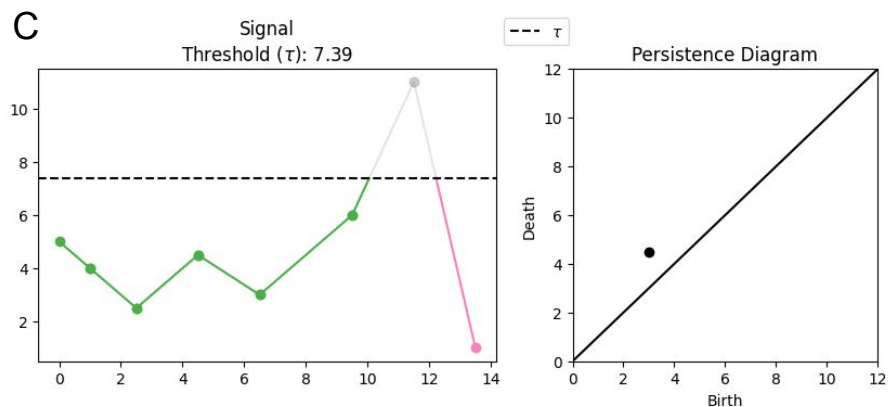
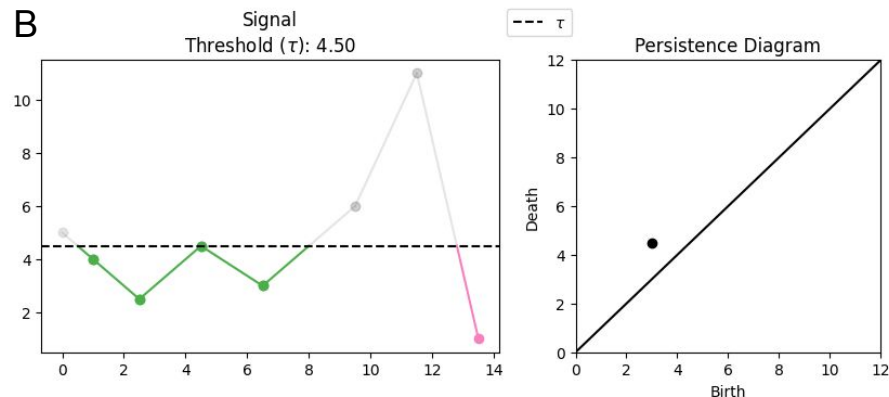
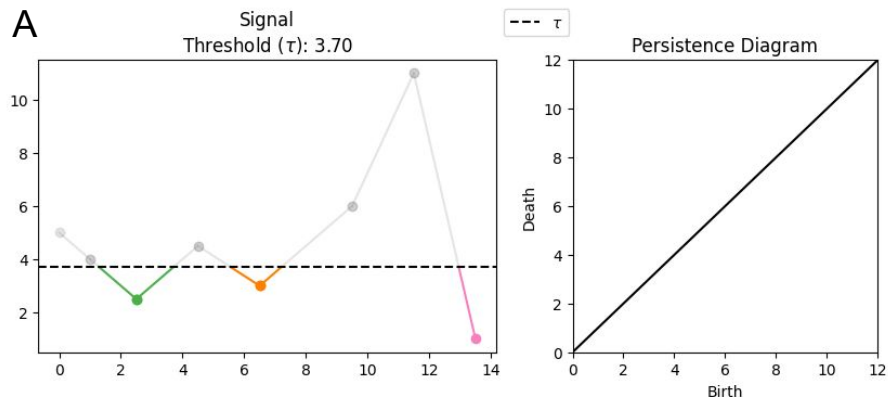
- Topological Data Analysis (TDA) Background.
 - Persistent homology and its application to signals.
- Topological Signal Compression (TSC).
 - Brief mathematical justification and a quick example.
- Free-Spoken Digit Dataset (FSDD).
- Machine Learning.
- Counterfactual Lossy Compression Algorithms.
- Results.
- Theoretical Benefits of TSC.
 - Contrast TSC with counterfactual compression algorithms.
- Future Work.

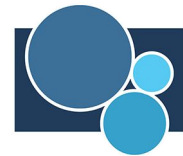


TDA Background

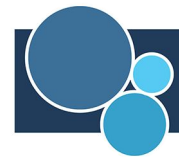


Persistent Homology and Persistence Diagrams



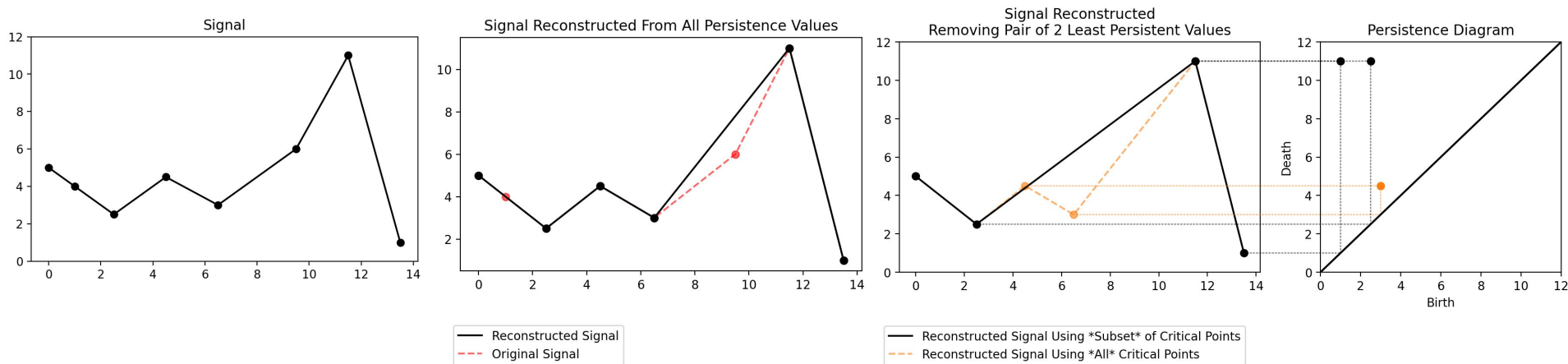


Topological Signal Compression (TSC)



Topological Signal Compression (TSC)

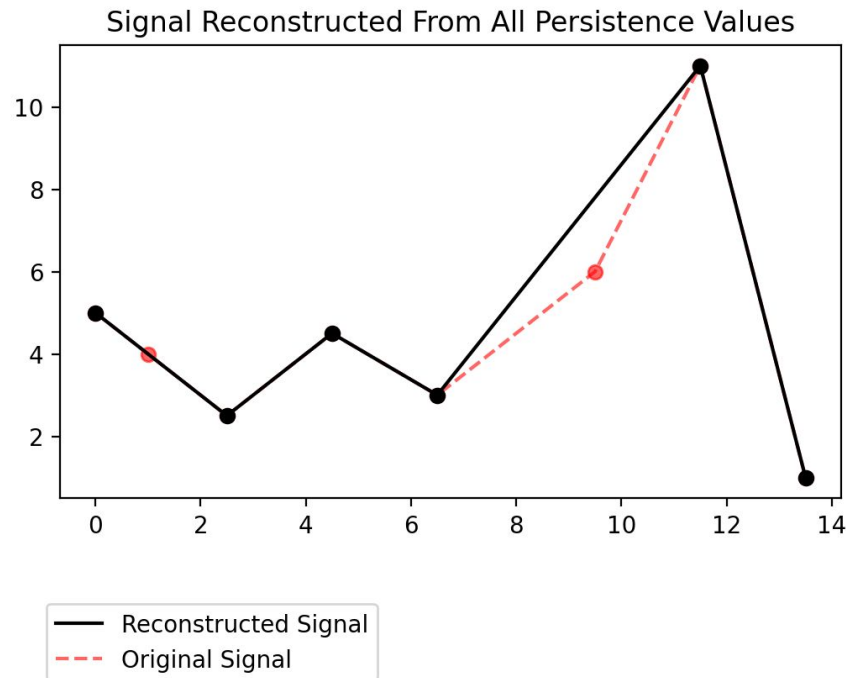
- Persistent homology thus gives us an *ordering* on our connected components. Dots on the persistence diagram that are closer to the diagonal die sooner after being born.
- TSC enables the reconstruction of the signal by *removing* the critical points (local mins and maxes) of least persistence, keeping the more prominent features of the signal.





Minimum Compression with TSC

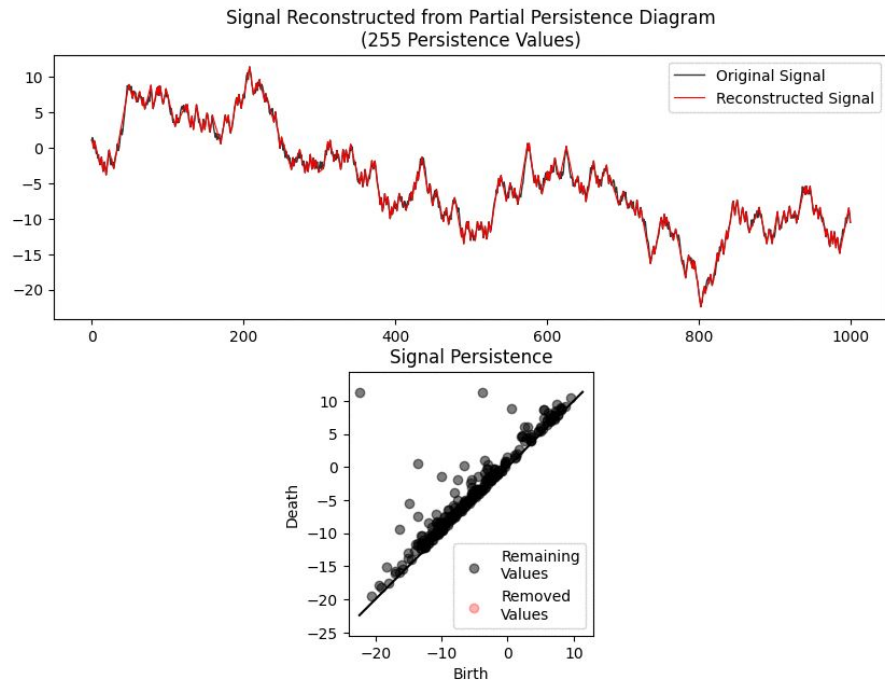
- In tracking the births and deaths of components, we only track the *critical points* in a signal.
- The minimum compression performed by TSC is to drop all non-critical points.

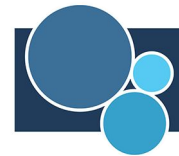




Mathematical Basis of TSC

- The *Morse Cancellation Lemma (MCL)* guarantees that the two values from the signal corresponding to the dot of lowest persistence are *horizontally adjacent* critical points in the time domain.
- This guarantees that *removing* this pair of points will not disrupt any other pair, thus preserving the critical point information over the rest of the signal.
- MCL can be applied iteratively to achieve arbitrary levels of compression, as demonstrated in the figure to the right.





Free-Spoken Digit Dataset (FSDD)



FSDD Summary

- Audio dataset produced by six male speakers consisting of recordings of spoken digits 0 through 9.
- 3000 examples recorded at 8 kHz, with each person recording 50 samples of each digit.



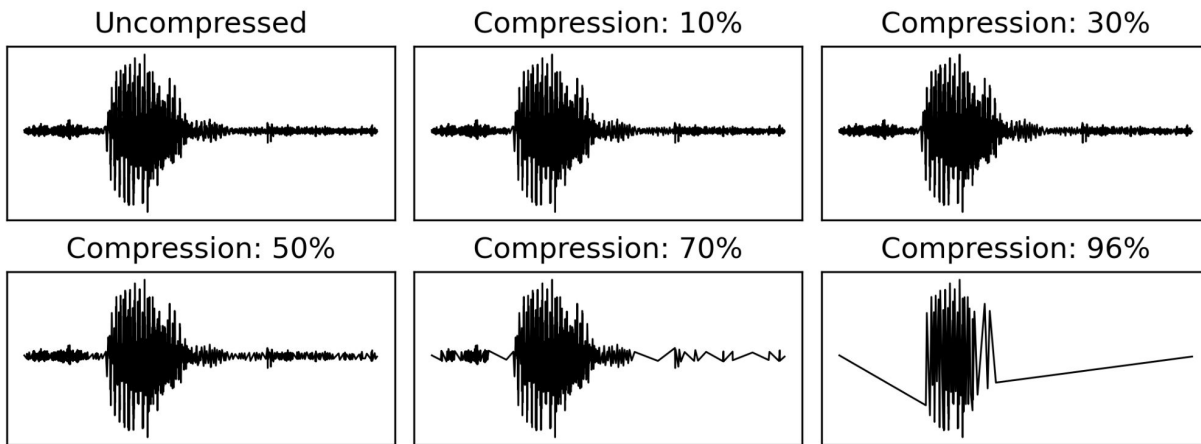
Machine Learning



Machine Learning on Compressed Data

- Repeat this cross-validated classification pipeline, instead building the MFCC features using increasingly compressed signals from FSDD.

Topological Signal Compression
Free-Spoken Digit Dataset (FSDD) Example Signal (Spoken Digit 6)





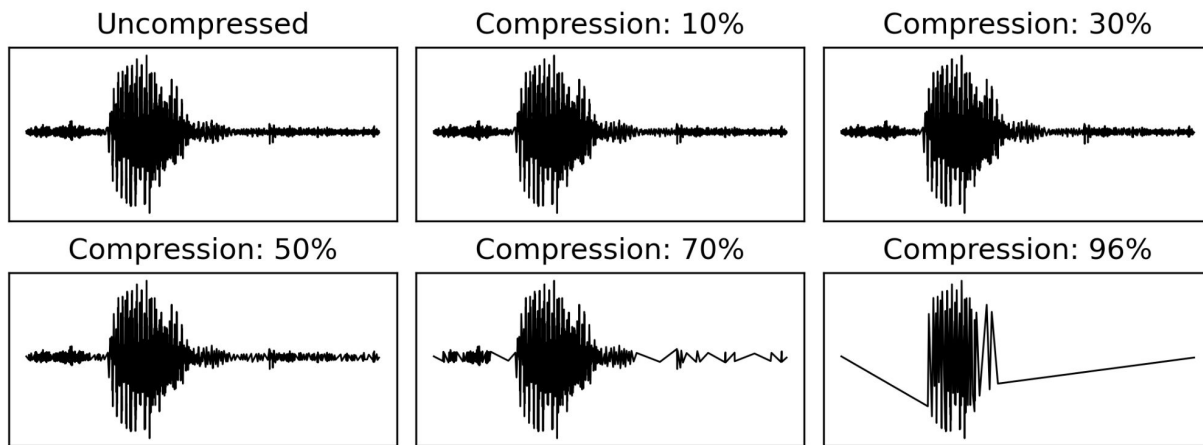
Counterfactual Lossy Compression Algorithms

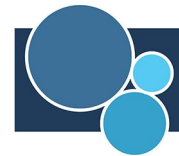


Our Algorithm: TSC

- Dropping data corresponding to data pairs with the lowest persistence values.
- More persistence values dropped $\Rightarrow$ more compression.

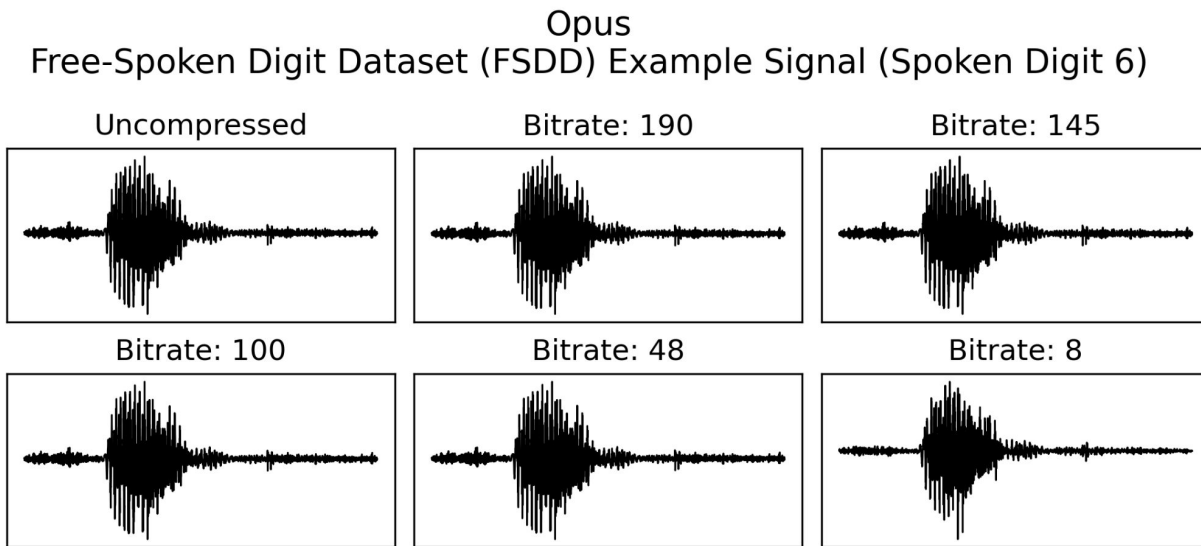
Topological Signal Compression
Free-Spoken Digit Dataset (FSDD) Example Signal (Spoken Digit 6)





Opus

- Codec designed exclusively for lossy audio compression.
- Lower bitrate $\Rightarrow$ more compression.

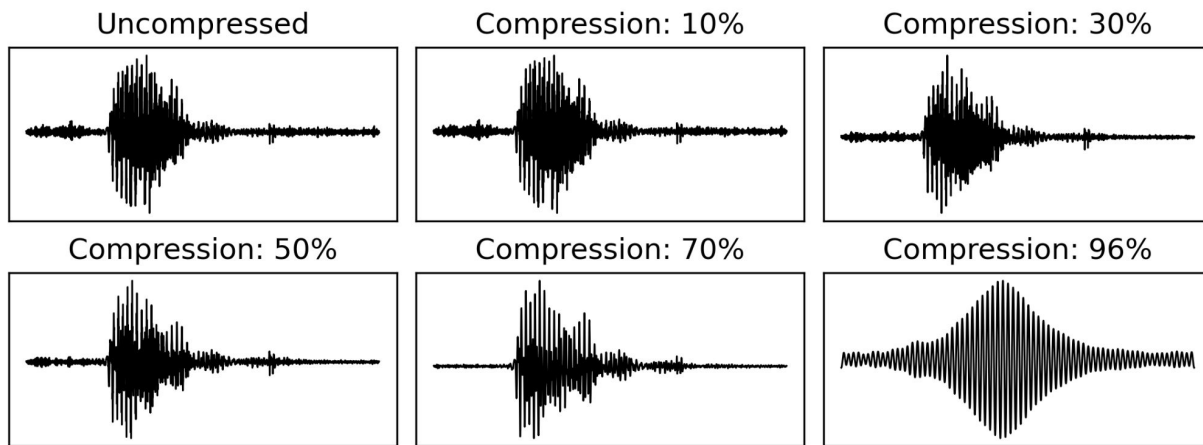


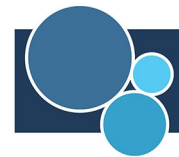


Discrete Fourier Transform (DFT)

- Represent a signal with its Fourier coefficients, sending only a subset of those coefficients to reconstruct the signal.
- More Fourier coefficients dropped $\Rightarrow$ more compression.

Discrete Fourier Transform
Free-Spoken Digit Dataset (FSDD) Example Signal (Spoken Digit 6)

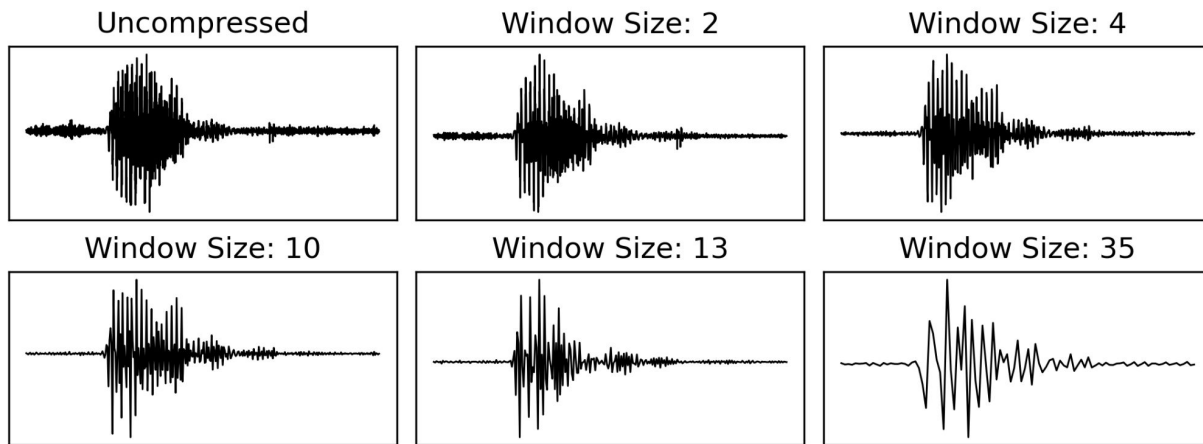


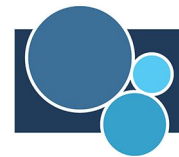


Piecewise Aggregate Approximation (PAA)

- Uniformly partitions a signal using a fixed window size and returns the mean function value within each window.
- Larger fixed window size $\Rightarrow$ more compression.

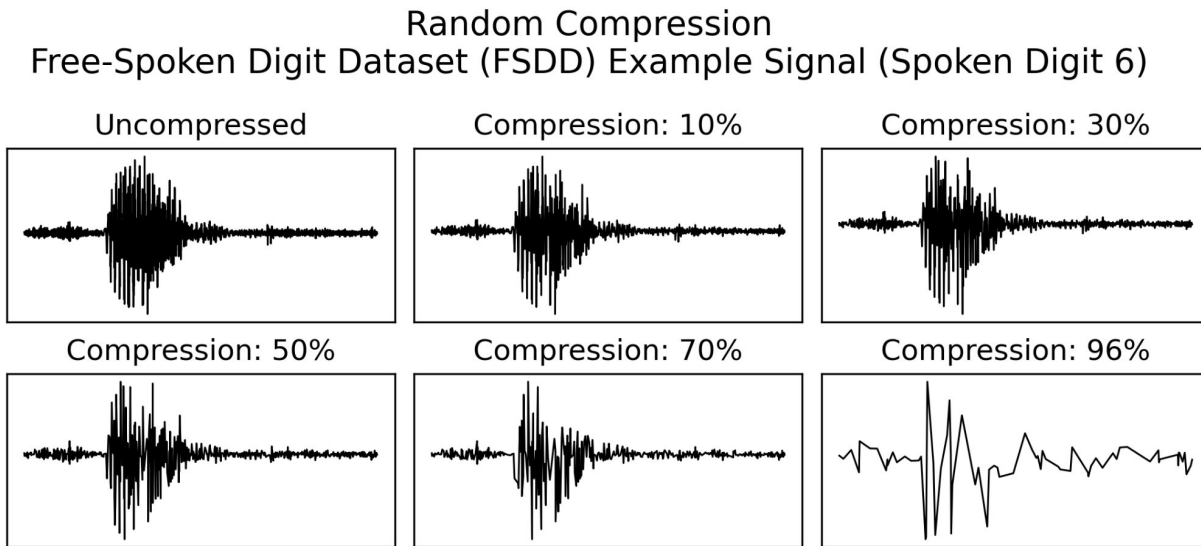
Piecewise Aggregate Approximation
Free-Spoken Digit Dataset (FSDD) Example Signal (Spoken Digit 6)





Random Compression

- Random sample (without replacement) of signal values.
- Less values sampled $\Rightarrow$ more compression.



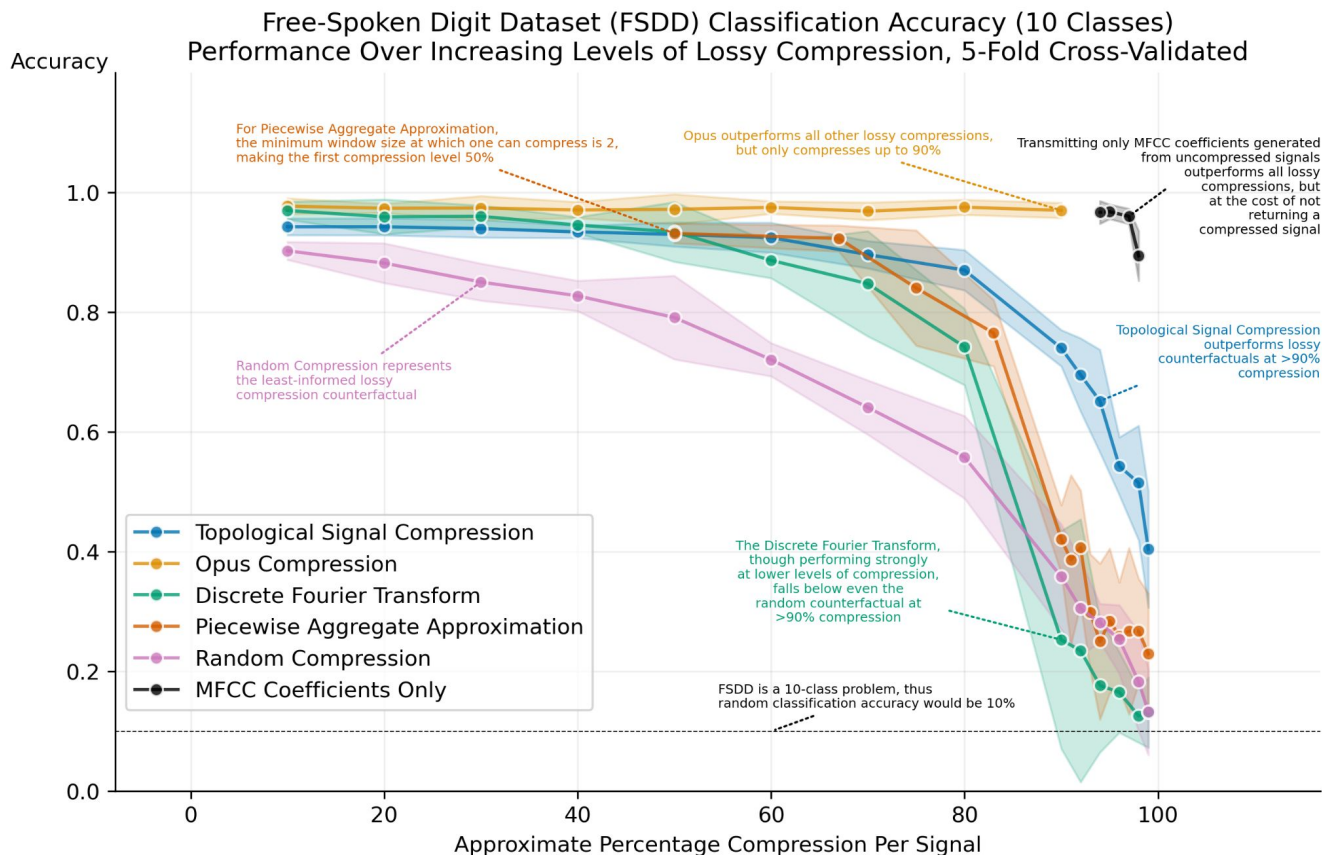


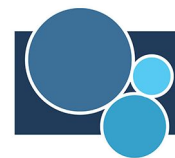
Results



ML Accuracy v. Compression Percentage

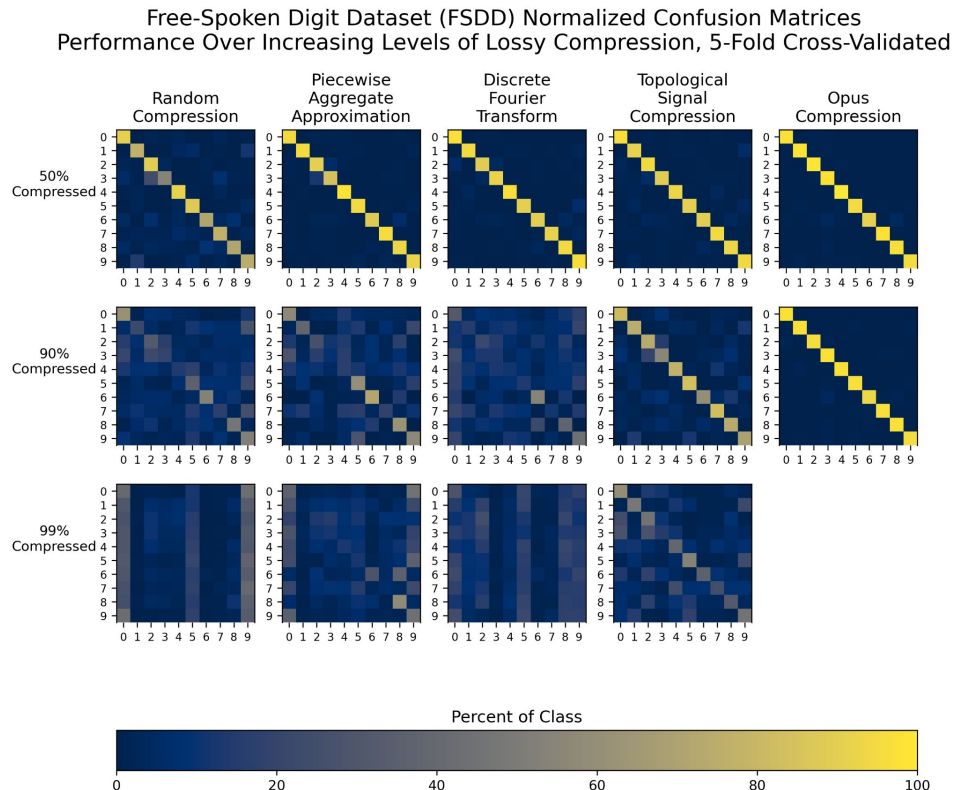
- Opus best, but only compresses up to 90% on average on FSDD.
- TSC outperforms other counterfactuals at >90%.





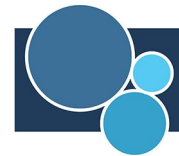
Confusion Matrices

- TSC maintains more consistent classification accuracy over each digit than the non-Opus counterfactual compression algorithms.



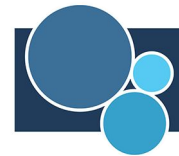


Theoretical Benefits of TSC



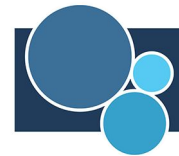
Generalizability

- Paper explores audio data, but TSC could be used on any other 1d signal.
 - Not the case for Opus.



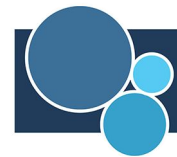
Interpretability

- A marginal compression with TSC creates a visually-interpretable change in the compressed signal.
 - Not the case for DFT, Opus.



Localized, On-the-Margin Compression

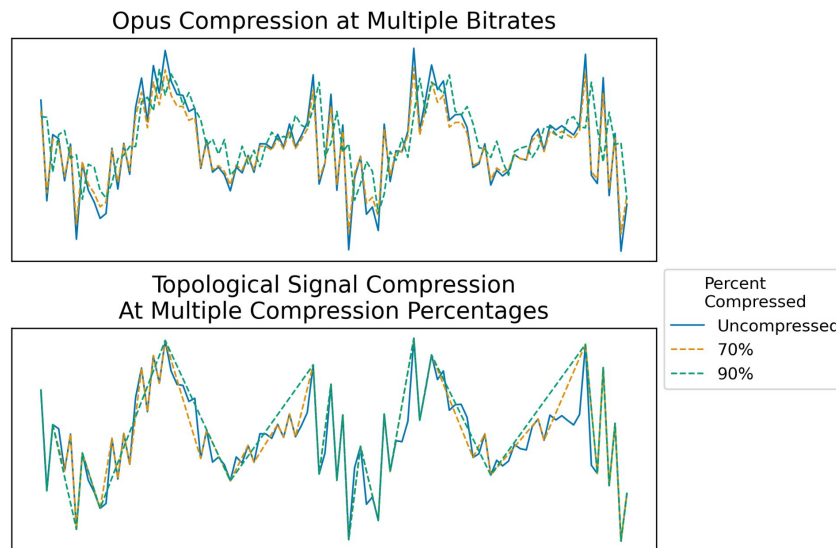
- Only TSC and DFT capable of byte-specific marginal compression.
 - Critical for using full bandwidth constraints.
 - Not the case for Opus and PAA.
- TSC only algorithm with *localized* marginal compression.
 - Not the case for DFT (or Opus, PAA).
- TSC likely more stable for downstream machine learning on variably-compressed signals.



Preserving Exact Values in Signals

- If exact signal values matter, TSC will preserve the exact critical points kept from a signal.
 - Not the case for Opus (see figure on right), DFT, or PAA.

Opus Compression vs. Topological Signal Compression
Free-Spoken Digit Dataset (FSDD)
Example Signal Subset (Spoken Digit 6)





Future Work



Future Work

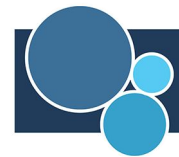
- Look at more modalities of data.
- More exploration into theoretical robustness of downstream ML to variably-compressed signals.



Thank You

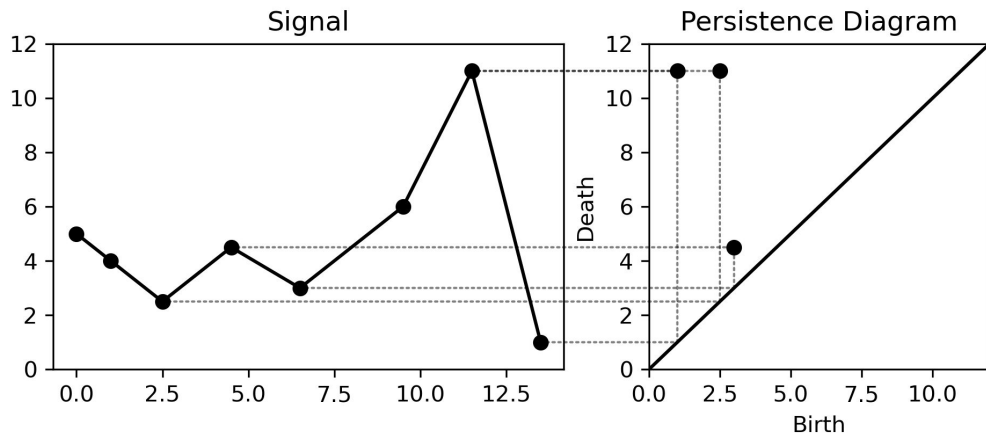


Appendix



Persistent Homology and Persistence Diagrams

- *Persistent Homology* tracks connected components as we sweep a horizontal line vertically from negative infinity to positive infinity.
- Components are born at local minima, and die at local maxima, destroying the more recently born component of the two components merging.
- The *Persistence Diagram* $D_0(f)$ plots the births and deaths of components as dots in the plane.
- The vertical distance of a dot above the 45 degree line $\text{birth} = \text{death}$ represents the *persistence* of a component.

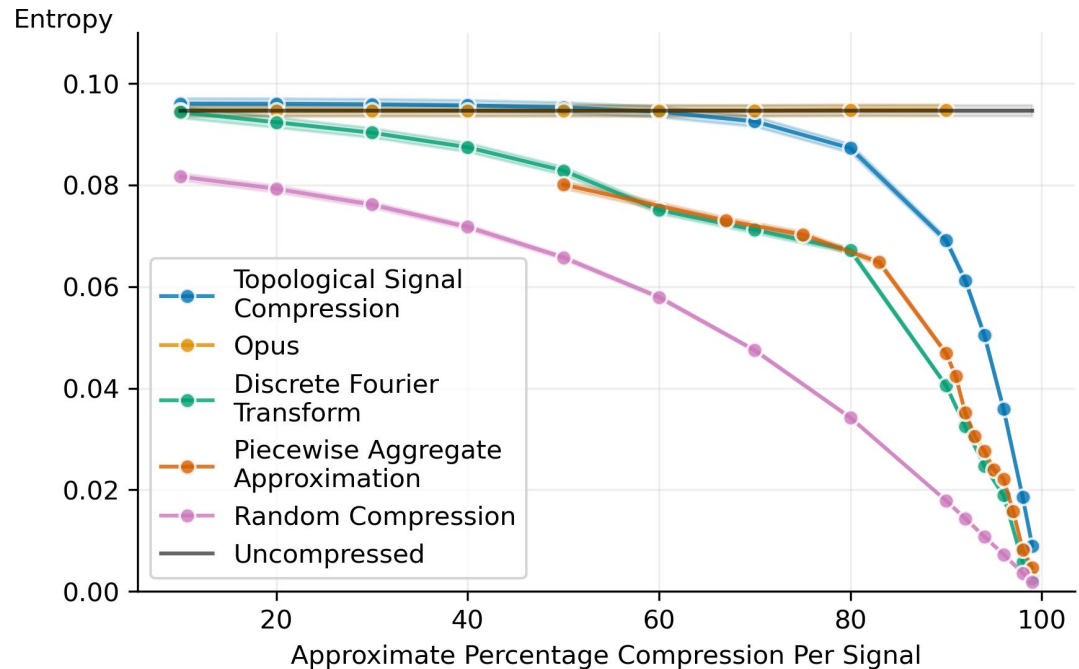




Signal Entropy v. Compression Percentage

- Entropy results comparable to ML results.

Free-Spoken Digit Dataset (FSDD)
Entropy Over Increasing Levels of Lossy Compression

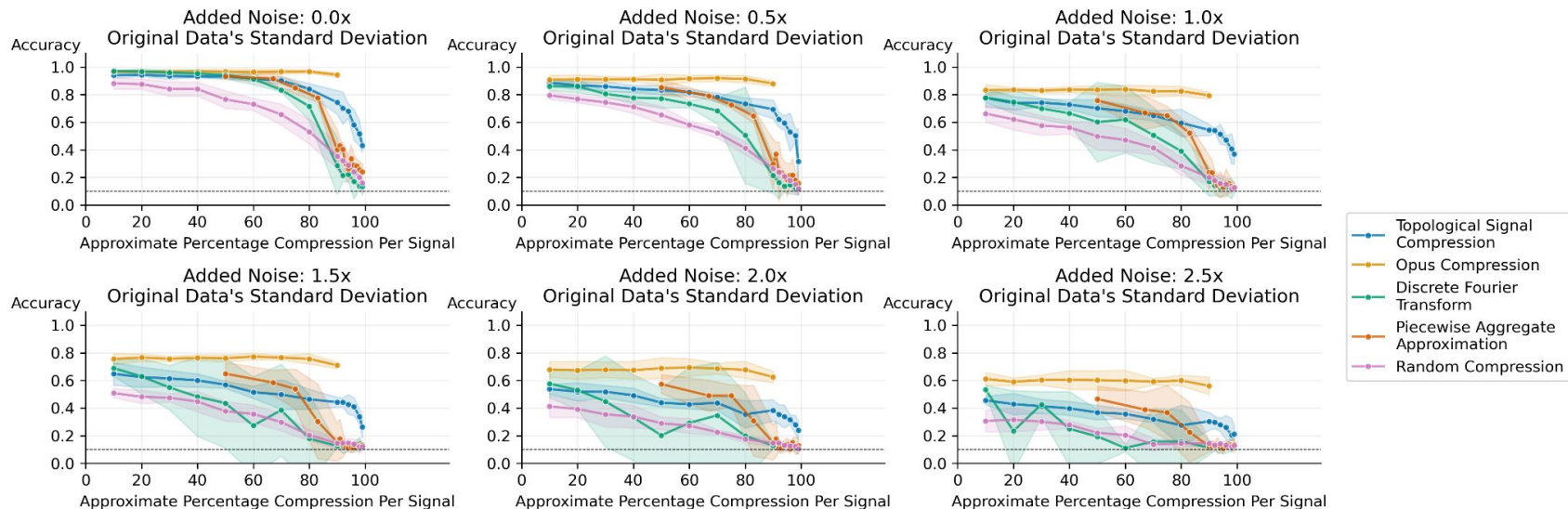




ML Accuracy v. Compression Percentage with Noise

- TSC's relative performance preserved at high compression levels with added noise.

Free-Spoken Digit Dataset (FSDD) Classification Accuracy (10 Classes)
With Increasing Levels of Added Gaussian Noise
Performance Over Increasing Levels of Lossy Compression, 5-Fold Cross-Validated

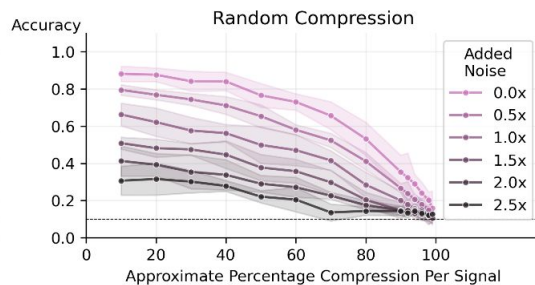
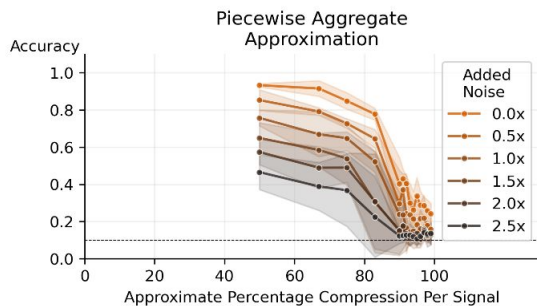
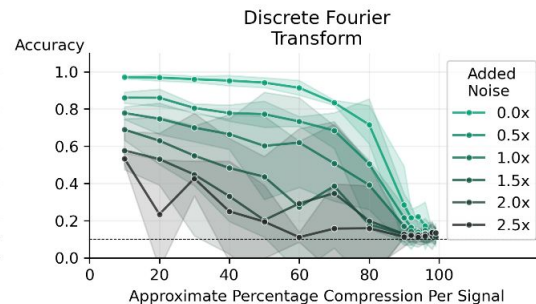
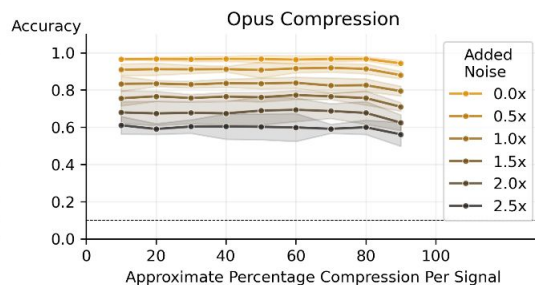
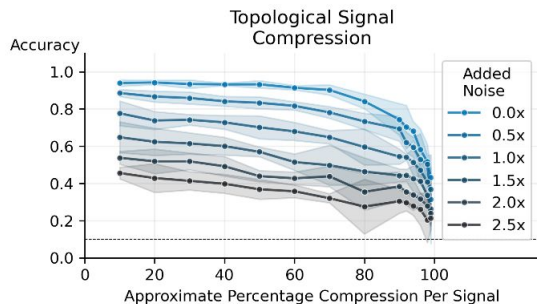




ML Accuracy v. Compression Percentage with Noise

- TSC's declining accuracy smooth over increasing noise.

Free-Spoken Digit Dataset (FSDD) Classification Accuracy (10 Classes)
With Increasing Levels of Added Gaussian Noise (Separated by Compression Type)
Performance Over Increasing Levels of Lossy Compression, 5-Fold Cross-Validated

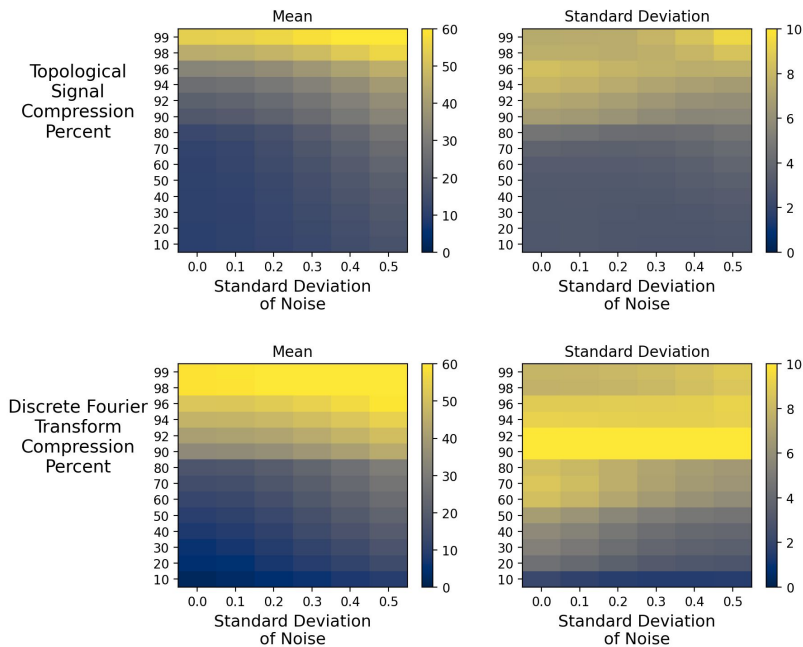




ML Accuracy v. Compression Percentage with Noise

- TSC's declining accuracy smooth over increasing noise.

Dynamic Time Warping Distance
Between Compressed Signals with Noise and Original Signals
(Mean and Standard Deviation)





Minimum Compression Percentage with TSC

- Values greater than 100% result from needing to store $(t, f(t))$ values with TSC instead of just $f(t)$ values.

